CSA5108 Cryptography and Network security

Exp 23 : Counter Mode (CTR) Using S-DES
Code:

```
#include <stdio.h>

int E(int x, int k) {
   return x ^ k;
}

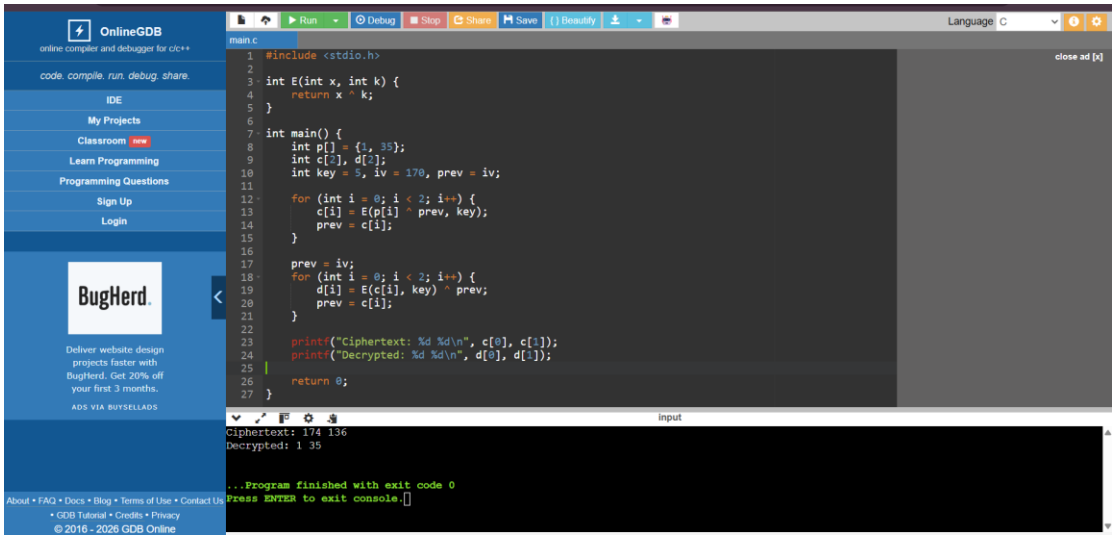
int main() {
   int p[] = {1, 35};
   int c[2], d[2];
   int key = 5, iv = 170, prev = iv;

   for (int i = 0; i < 2; i++) {
      c[i] = E(p[i] ^ prev, key);
      prev = c[i];
   }

   prev = iv;
   for (int i = 0; i < 2; i++) {
      d[i] = E(c[i], key) ^ prev;
      prev = c[i];
   }

   printf("Ciphertext: %d %d\n", c[0], c[1]);
   printf("Decrypted: %d %d\n", d[0], d[1]);

   return 0;
}
```


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Language C
main.c
close ad [x]
1 #include <stdio.h>
2
3 int E(int x, int k) {
4 return x ^ k;
5 }
6
7 int main() {
8 int p[] = {1, 35};
9 int c[2], d[2];
10 int key = 5, iv = 170, prev = iv;
11
12 for (int i = 0; i < 2; i++) {
13 c[i] = E(p[i] ^ prev, key);
14 prev = c[i];
15 }
16
17 prev = iv;
18 for (int i = 0; i < 2; i++) {
19 d[i] = E(c[i], key) ^ prev;
20 prev = c[i];
21 }
22
23 printf("Ciphertext: %d %d\n", c[0], c[1]);
24 printf("Decrypted: %d %d\n", d[0], d[1]);
25
26 return 0;
27 }
input
Ciphertext: 174 136
Decrypted: 1 35
...Program finished with exit code 0
Press ENTER to exit console.